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Identifying the subject in psychological research

Human sources for psychological generalizations

Making psychological knowledge claims, we saw in the last chapter, involves an ordered set of attributions attached to a particular kind of subject. The significance of these attributions depends in a crucial way on the nature of the subject whose attributes they are claimed to be. If the subject is an individual consciousness, we get a very different kind of psychology than if the subject is a population of organisms. The gradual predominance of the latter type of subject was seen to depend on the interest of psychologists in a type of knowledge that was applicable to populations outside the laboratory but that was as abstract as the generalizations of the physical sciences. This meant working with artificially defined collective subjects in the research situation and then imposing these definitions on people outside the laboratory.

In other words, psychological research on populations had a tendency to replace the social categories that defined populations in real life with populations defined in terms of nonsocial categories. American psychology aimed to be a socially relevant science, but not a social science. Its approach was to be that of a natural science, although its ultimate field of application was to be found among members of real societies.¹ Thus, research on socially defined populations came to be regarded as “applied” research, whereas “basic” research worked with abstract populations. In the extreme case the populations need not even be human. In fact, white rats fulfilled the role of a population of completely abstract organisms more adequately than any human population could hope to do.²

But where human populations were retained, the making of appropriately general knowledge claims faced a problem: Each empirical investi-

gation was a historically unique event, involving the interaction of particular participants at a certain place and time, yet some product of this interaction had to be presented as valid independently of these special historical conditions.³ On one level a psychological experiment is just another social situation in which specific individuals interact at a certain time and place and under particular circumstances. To advance even a mildly convincing claim that, unlike ordinary social situations, this situation yielded some special kind of knowledge, considered ahistorically valid and transsituationally relevant, one has to be able to point to special features of the experimental situation. These special features might be sought in the nature of the participants, the nature of their activity, and the nature of the circumstances to which they were exposed.

To start with the last of these possibilities, one could try to control the circumstances by reducing their apparent complexity, ambiguity, irregularity, or meaningfulness. Instead of having the participants interact under normal life circumstances, one could expose them to pure tones, reduced visual fields, regular stimulus series, meaningless syllables, and so on. In the "results" section of one's research report one could then present these special features of the experimental situation as though they were the only features of the situation that were relevant to one's knowledge claim. It seems that in the early days of experimental psychology claims to scientificity rested very largely on the factor of stimulus control. This could lead to somewhat ludicrous situations, because the reliance on this factor was easily transferred to the (brass) instruments that were the means used to actualize it. Thus we find that early attempts by American experimentalists to enter the field of clinical research were bedeviled by their misplaced attachment to their hardware.⁴

The second special feature of the experimental situation involved the nature of the activity to which the participants were limited. A fixed structure was imposed on the activity of experimental participants. That structure segmented their activity into artificial units that could be counted and aggregated, thus leading to the production of quantitative data that permitted a precise comparison of various effects.⁵

Discussions that have either extolled the merits of the experimental method or criticized it have generally revolved around one or both of these special features. Because psychological experimentation in principle shares these features with experimentation in the natural sciences, such discussions tend to reinforce the view that doing psychological experiments means practicing natural science. But in the reporting of psychological experiments something else happens, namely, the attribution of the data to a human data source. The nature of this source has as much to do with the kind of knowledge claim made in the experimental report as have the other two factors. Although the pervasive belief in a natural-science model of

psychological experimentation has generally prevented any discussion of this aspect, it is of profound importance for an understanding of the historical development of psychological investigation as social practice.

Consider the difference between a literary report on a human interaction and its outcome and the standard way of reporting when that interaction is a psychological experiment. One crucial difference between these two kinds of reports is to be found in the way the participants in the interaction are referred to. In the literary report they are likely to be identified by highly diversified individual and historical information, but in the research report any reference to the participants will be stereotyped and relatively poor in information. The claim that the outcome of the experimental interaction is not to be taken as a historically unique product but as a "finding" of potentially universal validity depends as much on a special way of describing and constituting the human data source as it does on other features of the experimental situation.

We therefore need to distinguish between the *identity* of the research subjects and their *identification* in the experimental report. All human participants in experimental situations necessarily have some identity as individuals and as members of certain social categories. They will also be referred to in some way in the published presentation and discussion of the research results. This experimental identification may or may not coincide with their normal personal or social identity. In this section of experimental reports, what is omitted may be just as significant as what is mentioned explicitly. Readily available information about the identity of participants may be omitted from the results section of the experimental reports because it is considered to be irrelevant to the scientific status of the knowledge product.

Another necessary conceptual distinction is that between the identity that participants have outside the experimental situation and the identity they have within it. As noted, the former involves the unique personal identity of the individual and the identity conferred by membership of various social categories like those of age, profession, and educational status. But because experimental situations involve a division of functions, the published reports are also likely to identify the participants in terms of these functions, as either experimenters or subjects. These temporary identities, which begin and end with the experimental situation, are deliberately created in order to construct such a situation.

These distinctions lead to two sets of questions. First, we will try to establish what kinds of extraexperimental identities were most common among those who participated in psychological research situations and ask what significance the various identities had for the knowledge claims that emerged. Second, we will examine the role of intraexperimental identities in the construction of research reports and their knowledge claims.

Social identity of research participants

Broadly speaking, there are two kinds of functions in the experimental situation for which the external social identity of the participants may be highly relevant. The first such function is that of the subject or data source, where the participant's social identity may be relevant because the goals of the investigation treat him or her as a representative of a particular socially defined group, such as male, female, or grade-three child. But there is also the function of scientific observation, reporting and recording of what was observed, which traditionally requires someone with an appropriate social identity. Experimental reports emanating from unqualified persons will not be taken seriously and will not be published in the recognized scientific literature. By the time experimental psychology makes its historical appearance the social identity of those functioning as experimenters and/or authors of experimental reports is on the whole unproblematical within rather narrow limits. The one who vouches for the reliability and accuracy of the observations and for the care with which they have been made must possess appropriate academic qualifications and experience. In other words, he or she must have a particular social identity.

The function of making observations that count as scientific is one that is common to the experimental situation in psychology and in other areas of experimental natural science. The function of the human data source is peculiar to psychology experiments and links them to the social sciences. But there is no reason *in principle* why these two functions should not be carried out by the same person. There may be practical difficulties in combining the two roles and there may be specific reasons for separating them in particular investigations, but in fact many of the classical pioneering studies in experimental psychology combined them very successfully. One need only think of Fechner's work in psychophysics and Ebbinghaus's original work on memory to appreciate this fact. As we saw in chapter 2, the reasons that prompted the early experimentalists to initiate a division of labor in the psychological laboratory appear to have been of an essentially practical nature.

Thus it is entirely understandable that in a sample of three volumes of Wundt's journal *Philosophische Studien* for the period 1894–1896 about 70 percent of the published experimental studies feature academic psychologists in the role of subjects. (In the successor journal *Psychologische Studien*, this proportion even reaches 80 percent for the 1909–1911 period.) In most of these studies, the subject was required to make careful sensory discriminations in the presence of systematically varied external stimuli, and his social status was a guarantee that these difficult and tiring observations would have been made with due care. In this respect experimental psychology was no different from experimental physics or other natural

sciences that insisted that only observations made by persons of appropriate background and experience had scientific credibility. The difference lay in the way in which these observational data were *interpreted*: In the case of physics they were interpreted in terms of events in the observed object; in the case of psychology, in terms of events in the observing subject.

For this model of psychological experimentation, prominent in the early years of the discipline, the social identity of experimental subjects was important in the same way that social identity was important in experimental science in general – namely, as a guarantee of the credibility of the observational data. But as we saw in chapter 4, there were other models of psychological investigation extant at an early stage, and for these the social identity of the participants was affected by additional considerations.

One departure from the Leipzig model involved the investigation of psychological processes in representatives of populations with a social identity that was felt to be psychologically interesting. This kind of investigation was in the tradition of clinical experimentation described in chapter 4. Early examples involved categories of persons with special abilities or deficiencies – somnambulists, blind or intellectually deficient persons as well as musically gifted persons and lightning calculators. In all these cases the individual functioned as a subject of psychological investigation much as someone with an unusual physiological or pathological condition might function as a subject for medical investigation. The difference was that whereas medical science already selected its cases by imposing its own categories, early experimental psychology essentially adopted culturally established categories and treated them as psychological (or biological) categories. In this model of psychological research, therefore, the social identity of individuals in the subject role was the crucial factor that pushed them into that role and kept it separate from the investigator's role. Being an appropriate candidate for medical or psychological investigation went with membership in certain social categories. In these cases the role that the subject played in the experimental situation was essentially an extension or elaboration of the role that went with his or her social identity.

Another early departure from the scientific norm of the psychologist–subject–observer occurred in studies that amounted to a kind of psychological census taking.⁶ Such studies were essentially descriptive. Mental phenomena were not studied as processes but as isolated instances to be counted. They were not related to any context, and the individuals who reported them only acted as a kind of neutral medium through which these phenomena manifested themselves. In their recording of the incidence of various psychological phenomena, these investigations seemed to be modeled on quantitative natural history. Because the phenomena to be studied were not treated as processes to be analyzed, no special observational skill or experience was required from the data source. Thus, virtually any unimpaired adult with a certain level of literacy could act as a subject for

Table 6.1. *Background of those participating as subjects in published psychological studies*

	Academic psychologists	Undergraduates	School children	Noneducational
1894–1896	52	38	13	3
1909–1912	24	38	28	9
1924–1926	11	40	36	11
1934–1936	10	52	27	14

Note: See the appendix for notes on this table.

these investigations. The only requirement was that they be available in some numbers to permit a convincing counting of instances. Undergraduate students fitted this role perfectly.

Research students also appeared as experimental subjects in some numbers. However, they functioned as expert observers, exactly like fully qualified academic psychologists. Their use, therefore, introduces no new considerations into the present analysis.

Children, who constitute the last of the major categories of subjects for psychological research, do need special consideration. At least from the age of school entry onward, children were always a significant source of psychological data for American investigators. In the early years school children were prominent subjects for the practice of psychological research as a kind of quantitative natural history. Much of this work was inspired by G. Stanley Hall, as we have already noted. School children shared with undergraduates the advantage of an institutional context that made them conveniently available in socially classified groups and compelled their participation in the research process. But, unlike undergraduates, school children were also of great interest as potential objects of practical psychological intervention. They constituted by far the most important category of persons with special deficiencies and were therefore appropriate objects for clinical experimentation.

Table 6.1 presents some temporal trends in the use of different categories of research subjects. Certain general trends emerge rather clearly. Academic psychologists are at first the most important group of subjects for psychological research and then show a progressive decline. This is undoubtedly a reflection of the fate that befell the Wundtian model of psychological research practice, as noted in chapter 4. In view of this decline it is perhaps surprising to find that the use of subjects from outside educational institutions increased only very gradually during the period covered. Moreover, during the 1920s and 1930s most of the studies employing this category of subject come from a single source, the *Journal of Applied Psychology*. The core of the discipline remained uninterested in most of

the human population as a source of research subjects. When experimental psychologists ceased to take the subject role themselves they turned to animals, to children, or to undergraduates as the three alternatives. Among these, undergraduates were clearly the favorites. They had always been a major source of data in American psychological research, and in the period after World War I they reach a position of predominance which they have never lost.⁷

There was also a change in the way in which students were used. In the early period it was quite unusual to publish studies in which undergraduates were the *only* source of experimental subjects. As a rule, undergraduates would only appear in studies that had also used the services of other kinds of subjects. Comparisons of results from subjects with different backgrounds were not uncommon. This continued to be the predominant pattern up to the second decade of the twentieth century. However, in the period after World War I studies based only on undergraduate subjects became much more frequent and outnumbered those studies in which both undergraduates and other categories of subject were employed. By the 1930s, two-thirds of the studies that used undergraduates used them as the sole source of psychological data. The role of undergraduates as research subjects seems to have changed. At first, their extraexperimental identity still had some significance for their research role, insofar as their reactions were compared with those of subjects with a different identity. But, increasingly, they came to be used as a convenient source of abstract experimental subjects whose extraexperimental identity was considered to be irrelevant in the context of psychological experimentation.

Even in the days when it was common for subjects with a different extraexperimental identity to be compared with each other, the investigators did not usually think in terms of social identities. The categories in which most early American psychologists thought about the objects of their research were derived from biology, not from sociology or history. So the comparison of different kinds of subjects was more likely to be interpreted in terms of quasi-biological categories, like age or innate endowment, than in terms of history or culture. Gradually, the link to biological categories became more and more remote, as psychology developed its own abstract categories for the description of human action. The further this process developed the more completely substitutable did experimental subjects become. Individuals simply became the media through which abstract laws of behavior expressed themselves. Because the abstract laws and not their individual carriers were the primary objects of investigation, the social and even the biological identities of experimental subjects became theoretically irrelevant. The choice of subjects could be governed solely by practical convenience. On that criterion no group of human subjects could compete with college students.

The rhetoric of experimental identities

Published research reports reflect certain decisions that were taken about how to refer to the human source of the data. The author of such a report has the alternative of referring to such sources either by giving them their ordinary social identities, such as boys or girls, or by giving them special experimental identities, such as subjects or observers. This choice is not made randomly but depends on prevailing norms and conventions among the community of investigators. It is likely that these norms reflect prevailing conceptions of psychological knowledge.

One might think that the choice of how one refers to the human sources of one's data is a relatively trivial matter. This appearance of triviality is however the product of a situation in which the norms governing such matters have simply become hallowed by long usage, so that the awareness of their implications has been lost. When we look at the historical development of these norms we come across issues that are far from trivial.

In the first place, the history of experimental psychology is marked by a long period of indecision about the appropriate way to refer to the experimental identity of those who functioned as the source of data. In the very early days of the discipline a number of terms were in use (see chapter 2), but in the English-language literature two terms quickly overshadowed all the others: "subject" and "observer." By the end of the nineteenth century the former was being used in about half of all published reports, and the latter in about a quarter. But then it took another half-century for the use of observer to dwindle to a negligible level. Clearly, the connotations of these terms were not a matter of indifference at the time but presented authors and editors with a real choice.

In Table 6.2 the identification of the human data source as either subject or observer is traced over four time periods. Although a few studies employ both terms, most opt for one or the other. In general, the term subject has an early preponderance, which increases very gradually over the period under review. It is as though the rival term observer were fighting a determined rearguard action. The main center of resistance appears to be concentrated around the *American Journal of Psychology*. But this is not apparent until after the turn of the century. Between 1895 and 1910 a major swing from subject to observer takes place in this journal whose countertrend remains visible until the 1930s. It seems most likely that the original switch to the term observer in preference to subject was linked to the emergence of a more self-conscious kind of introspectionism, which presented itself as "systematic experimental introspection" (see chapter 3). In America, Titchener was the leading figure in this development, and the *American Journal of Psychology* became its main publication channel. This defined the role of the psychological data source entirely in terms of

Table 6.2. *Percentage of empirical studies identifying the human data source as “subject” and as “observer”*

	Subject				Observer			
	1894–1896	1909–1911	1924–1926	1934–1936	1894–1896	1909–1911	1924–1926	1934–1936
<i>American Journal of Psychology</i>	69	25	49	55	38	72	56	45
<i>Psychological Review</i> ^a	68	75			26	33		
<i>Psychological Monographs</i>	— ^b	82	61	73	—	29	43	8
<i>Journal of Experimental Psychology</i>	—	—	92	88	—	—	12	20
<i>Journal of Applied Psychology</i>	—	—	71	35	—	—	0	0
<i>Journal of Educational Psychology</i> ^c	—	36	42	46	—	12	0	0

^aCoding discontinued for third and fourth period because of paucity of empirical studies.

^bDashes indicate periods before journal commenced publication.

^cPublication began 1910; hence first three volumes coded are 1910–1912.

introspective observation, whereas earlier practice had been much less restrictive. Not surprisingly, it also favored research areas whose content gave the most scope for the working out of its methodological program.

It seems that as a result of this development, and the subsequent reaction of the behaviorists, subject and observer acquired some ideological significance in terms of the internal divisions within American psychology. In the 1920s these terms were being pressed into service by rival schools of experimental practice. By 1929 one well-known experimental psychologist⁸ considered the time to be ripe to urge the dropping of the term observer from psychological discourse. After all, "in many contemporary lines of psychological investigation the so-called 'observer' does no observing." This led to a prompt reply (in the *American Journal of Psychology*, of course) by a prominent representative of a different point of view,⁹ who, while no longer defending Titchenerian introspectionism, objects to a "cult of objectivism" that refuses to recognize the special nature of those experiments "where the organism enters the scene as *agent*" (emphasis in the original). Further controversy¹⁰ did not resolve the divergence of perspectives.

Although the ideological connotations of the terms subject and observer were probably salient only for a minority of experimentalists, the failure of the discipline to establish uniformity of usage even after several decades is a reflection of a continuing division of opinion about the essential function of the human data source. There appears to have been a general feeling that the data source ought to be identified by reference to its essential function in the experimental situation. But what was the essential function? The "objectivists" clearly felt that it consisted in providing material for the experimenter's observation and use, and the term subject had by long usage acquired this kind of meaning (the medical prehistory of the term had been forgotten, although it was preserved in practice). The minority who resisted this trend saw the essential function of the human data source somewhere else, namely, in its competence to comprehend the experimental task and act appropriately. This latter interpretation was of course quite close to Wundt's original conception of the role of the human data source in psychological experiments. As we have noted, experimental psychology began life with several different conceptions of its practice. One of these was in retreat almost from the start, although complete uniformity of practice does not appear to have been reached during the period under review here.

Both subject and observer refer to a strictly intraexperimental identity. Persons only have these identities insofar as they play a particular role within the social situation constituting a psychological investigation. However, these same persons do not thereby lose their ordinary social identities and may still be identified by the latter in experimental reports. The frequency of such "extraexperimental" identifications in published empirical

Table 6.3. *Percentage of empirical studies identifying the human data source by an extraexperimental identity*

	1894–1896	1909–1911	1924–1926	1934–1936
<i>American Journal of Psychology</i>	19	11	5	20
<i>Psychological Review</i> ^a	13	18		
<i>Psychological Monographs</i>	— ^b	6	46	56
<i>Journal of Experimental Psychology</i>	—	—	28	7
<i>Journal of Educational Psychology</i> ^c	—	60	62	63
<i>Journal of Applied Psychology</i>	—	—	33	65

^aCoding discontinued for third and fourth period because of paucity of empirical studies.

^bDashes indicate periods before journal commenced publication.

^cPublication began 1910; hence first three volumes coded are 1910–1912.

studies is shown in Table 6.3. Most commonly these extraexperimental identifications refer to “children” and “pupils,” less commonly to “boys” and “girls,” and occasionally to “students” and to “men” and women.”

A clear difference emerges between journals devoted to “basic” experimental research (*American Journal of Psychology* and *Journal of Experimental Psychology*) and those more concerned with “applied” knowledge (*Journal of Educational Psychology* and *Journal of Applied Psychology*).¹¹ Subjects of psychological research were, on the whole, more likely to be identified by an extraexperimental identity if the investigation was intended to have some direct relevance outside the investigative context. In experimental work devoted to abstract psychological knowledge, however, the human source of the data was very likely to be identified only by an intraexperimental identity.

It is also the case that studies published in the “applied” journals were, on the whole, more likely to be based on the use of subjects with a non-academic background, either children or adults. In the *Journal of Applied Psychology* and the *Journal of Educational Psychology* the proportion of studies employing such subjects ranged between 50 and 95 percent for different periods, but in the *American Journal of Psychology* and the *Journal of Experimental Psychology* the range was between 11 and 37 percent. The research published in the latter two journals was more likely to be based on student subjects or, in the early period, on psychologists acting as subjects.

In general, the research literature that is based on the preferential use of undergraduates is most likely to identify its data source by an intraex-

perimental identity, whereas that which is based on the use of children or nonacademic adults is more likely to retain extraexperimental identities in reporting research results. An overall comparison for all six journals and all time periods covered in Tables 6.1, 6.2, and 6.3 indicated that children were five times more likely to be referred to by their common extraexperimental identities than were college students.

Why were children and nonacademic adults much more likely to retain their normal identity than were college students when they participated in psychological investigations? The conferral of particular identities in social situations depends on the perceived relevance of those identities for the social situation. There is no reason to suppose that the social situation that we know as a psychological experiment will be any different from other social situations in this respect. Whatever identities are conferred on the participants in the experimental situation will be the ones that are most relevant to the construal of that situation by those who report on it. If children often retained their extraexperimental identity in the experimentation situation whereas undergraduates were usually given a special identity, this must mean that the child's ordinary identity was considered to be experimentally relevant in a way that the undergraduate's was not.

In other words, investigations involving children were usually regarded as providing information specifically about children, whereas most investigations involving college students or academics were not regarded as providing information specific to the psychology of college students or academics. Sometimes the educational status of the college students who served as subjects was considered relevant to the purpose of the research, and in these cases we do find them referred to as "students."¹²

Conversely, the tendency to refer also to children as "subjects" increased steadily during the period covered. This may have been part of a more general tendency to emphasize the scientific pretensions of the discipline, pretensions that demanded that psychological research be concerned with abstract relationships rather than with people.

Experimental psychologists were concerned to establish a kind of knowledge about human beings that would be ahistorical and universal. Yet to obtain this knowledge they had to work with specific, historically defined human data sources, and they had to extract their data from these sources in investigative situations that were also historically specific. To cope with this paradox, psychologists commonly employed certain rhetorical devices when reporting their data. It became customary to emphasize the experimental identity of human data sources at the expense of their ordinary personal and social identity. Thus, their actions, as reflected in the "experimental results," were not attributed to them as historical individuals but to them as incumbents of a special experimental role. Where experimental results are presented as activity attributed to persons who have no identity apart from their identity as experimental subjects, the implication

is that no other identity is relevant in the experimental context. The experiment and its results stand apart from the sociohistorical context in which the rest of human life is embedded. It appears to lead straight to human universals, not infrequently identified with biological universals.

A further development that enhances this effect involves not attributing the experimental results to specific individuals at all, not even individuals that only have an experimental identity. Experimental results are then presented as the attributes of experimental groups whose anonymous members have no individual existence in the experimental report. This practice, discussed in chapter 5, can now be seen as having the additional function of enhancing the isolation of laboratory products from the personal and cultural reality that produced them.

The fact that the handling of the attribution problem in the presentation of experimental results involves rhetorical considerations does not mean that the knowledge claims implied in this presentation are necessarily false. It only means that they have not been empirically secured, that the issue remains wide open. Published reports of experimental results cannot avoid the task of establishing the claim that what is being reported can justifiably count as "scientific" information. For a discipline that took "scientific" to imply reference to some universal truth beyond individuality, history, and local meanings, the establishment of claims to being scientific frequently depended on appropriate manipulation of the identity of the sources to which data were attributed. With few exceptions, claims to universality were not grounded empirically but were established by fiat. The role of the experimental report was, among other things, to create the illusion of empiricism. By the way the experimental results were presented, the illusion was created that they were not really the product of a social interaction among certain human personalities in historical time, but that they were the manifestation of abstract transpersonal and transhistorical processes. As experimental psychology progressed, its published reports dealt more and more with abstractions and became less like accounts of particular investigations.

Marketable methods

Education as psychology's primary market

Thus far, we have traced the development of some of psychology's investigative practices in terms of the general knowledge interests pursued by psychologists. This may be adequate for the more general features of these practices but when we turn to more specific features it becomes necessary to take into account a broader social context. The fact is that almost from the beginning of the twentieth century psychology ceased to be a purely academic discipline and began to market its products in the outside world. That meant that the requirements of its potential market were able to influence the direction in which psychology's investigative practices were likely to develop. Practices that were useful in the construction of specific marketable products were likely to receive a boost, whereas practices that lacked this capacity were henceforth placed under a handicap.

Of course, the requirements of the market did not act on a passive discipline. Not only did many psychologists actively court practical application, but these efforts would have led nowhere if the discipline did not have at its disposal certain techniques that lent themselves to the development of a socially useful product. In the present chapter we will analyze two such techniques in terms of their significance for a socially relevant investigative practice: the Galtonian approach to individual differences and the experimental use of treatment groups. Both of these methodological innovations assumed enormous importance in the subsequent development of modern psychology, but it is doubtful whether this would have been the case had they not played a crucial role in the constitution of psychology as a socially relevant discipline.

However, if there is to be any real analysis of the social significance of these techniques, the category of social relevance will have to be brought